Barcode Inspection and Jidoka Demonstration

A desktop-scale PLC and industrial-camera project for barcode-based quality decisions

> **Project in brief.** This project demonstrates how an industrial camera and PLC can check barcode identity, indicate OK/NG results, handle no-read and communication-error conditions, and present the process through a simulated HMI.

1. Project overview

The project recreates a barcode inspection station in a desktop setup. A Keyence SR-X100W barcode reader is triggered from a PLC input. The first five successfully read barcodes are stored in a list in the PLC project. Subsequent reads are compared with that list: a matching barcode is treated as OK, while a different read barcode is treated as NG.

The purpose is to demonstrate a simple quality-checking and Jidoka-style response: the system makes a decision from the inspection result and provides a clear visual indication, while also identifying conditions in which a valid reading or camera communication is unavailable.

2. System components

Component	Role in the project
Siemens SIMATIC S7-1200 CPU 1215C DC/DC/DC	Runs the PLC logic for triggering, result handling, barcode comparison, status indication, error handling and reset.
Keyence SR-X100W	Reads the barcode and returns reading status and data to the PLC.
WinCC simulation on a PC	Provides the HMI view in software; no physical HMI panel was used.
Patlite K50LGRAL1YPQ	Provides visual and audible status indication: yellow at trigger, a brief buzzer at each read, green for OK and red for NG.

3. System block diagram

Operator trigger E0.0	→		↔	**Keyence SR-X100W barcode reader**

		Siemens S7-1200 PLC sequence, compare, status		
		↓		
Patlite indicator yellow / green / red / buzzer	←	**WinCC PC simulation operator display**	←	**Result and status data**

Conceptual diagram based on the project description. Physical wiring, exact network topology and complete I/O addressing are documented separately in the project's TIA Portal printout and I/O list.

4. How the process works

1. The operator presses the PLC trigger button at input E0.0. The system starts a camera reading cycle and turns on the yellow indication for the trigger event.
2. The camera reads the barcode and returns the reading status and barcode data to the PLC. The buzzer gives a short indication when a read event occurs.
3. The PLC stores the first five barcode strings as the reference list. Later barcode results are compared with the stored values.
4. If a read barcode matches a stored reference barcode, the PLC identifies the result as OK and turns on the green indication. If a barcode is read but does not match, the result is NG and the red indication is used.
5. If a reading does not complete within the configured 8-second timeout, the PLC records/indicates a timeout or no-read condition. A camera communication fault is handled as a COMERR condition.
6. The operator can use the reset button at input E0.1 to reset the relevant status and counters and prepare the demonstration for another run.

5. PLC program and supporting documentation

The TIA Portal LAD printout contains the PLC block and network diagrams. The visible program structure includes networks for link-status checking, communication-error handling, trigger/read sequencing, timeout, read completion, no-read, OK comparison, NG comparison, cleanup, communication recovery and reset. The I/O table and detailed LAD network pages are provided in the accompanying TIA Portal PDF rather than duplicated here.

The project steps document records the development path: reviewing hardware information, wiring and mounting, configuring PC and PROFINET settings, setting up the PLC and device data, implementing trigger/reset and barcode-to-string handling, adding LED/buzzer indications, and planning OK/NG/timeout logging and HMI display.

6. Development and demonstration

Work on the project began in October. The final test was carried out on 9 October and recorded on video. The video demonstrates the working test sequence; the accompanying TIA Portal PDF supplies the program/network and I/O-level reference material.

7. Scope and next steps

This is a desktop demonstration of a live-line inspection concept, not a complete production-line installation. The HMI is simulated in WinCC on a PC, and the system is intended to show the basic barcode decision and status-handling sequence.

Possible follow-on work includes implementing the planned external valid-barcode table and SQL Server logging, displaying barcode results and alarms in WinCC, and expanding the test record to cover successful reads, mismatches, no-read timeout, communication loss and recovery.

Prepared as a concise project introduction from the supplied project description, development-steps document and TIA Portal program printout.